

Token Economy Simulator  
Diagnostic Calculation Layer Specification  
Version 1.0 — Backend Implementation Document

Digital Communities and Territories Research Group

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# 1 General Principles

This document specifies the exact calculation logic for the diagnostic engine of the token economy simulator. It is intended for developer use and maps user inputs to failure mode risk assessments. All parameter IDs reference the `simulator_specification.xlsx` file.

The document is organized as follows:

1. Input pre-processing rules — how qualitative selections map to numerical variables
2. Derived variables — intermediate calculations required before failure mode assessment
3. Failure mode calculations (FM1–FM6) — formula, threshold, risk levels, output text, edge cases
4. Coherence check logic — cross-failure-mode contradiction detection
5. Overall risk score — aggregation and prioritized recommendations
6. Developer notes — computation order, unit conventions, precision rules

## 2 Input Pre-Processing Rules

These rules convert qualitative user selections into numerical values for use in formulas.

### 2.1 Monitoring Capacity $\gamma$ (Parameter 4.4)

| User selection                  | Numerical value |
|---------------------------------|-----------------|
| Low ( $\gamma \approx 0.1$ )    | $\gamma = 0.10$ |
| Medium ( $\gamma \approx 0.5$ ) | $\gamma = 0.50$ |
| High ( $\gamma \approx 0.9$ )   | $\gamma = 0.90$ |

### 2.2 Sanction Severity $S$ (Parameter 4.5)

$S$  is expressed as a fraction of the average participant token balance  $B_{avg}$  (computed in Section 3).

### 2.3 Temptation $T$ and Reward $R$ (for FM4)

$T$  and  $R$  are payoffs from game theory used in the monitoring condition  $\gamma S > T - R$ .

| User selection      | $S$ value                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| None                | $S = 0$                                                                                                                                                                                                      |
| Warning             | $S = 0.05 \times B_{avg}$                                                                                                                                                                                    |
| Token penalty       | $S = p \times B_{avg}$ , where $p$ is the penalty percentage from sub-input (default $p = 0.20$ )                                                                                                            |
| Exclusion           | $S = 1.00 \times B_{avg}$ (full effective balance loss)                                                                                                                                                      |
| Graduated sanctions | $S$ follows an escalation function: Step 1: $0.05 \times B_{avg}$ ; Step 2: $0.20 \times B_{avg}$ ; Step 3: $0.50 \times B_{avg}$ ; Step 4: $1.00 \times B_{avg}$ . Use Step 1 value for initial assessment. |

**Verification strength.** Each verification mechanism (parameter 1.5) maps to a verification strength score:

| Verification mechanism | Strength score |
|------------------------|----------------|
| Physical presence      | 0.95           |
| Smart contract         | 0.90           |
| Peer verification      | 0.80           |
| Third-party cert       | 0.75           |
| Self-reporting         | 0.10           |

**Temptation payoff.**

$$T = E_{contrib} \times (1 - v_s) \quad (1)$$

where  $E_{contrib} = E(t)/(N \times \phi)$  is the emission per active contributor per period and  $v_s$  is the verification strength score.

**Reward payoff.**

$$R = E_{contrib} - d \times \bar{P} \quad (2)$$

where  $d$  is average demand per participant (parameter 5.7) and  $\bar{P}$  is average token price (Section 3).

*Note:* if  $T - R \leq 0$ , condition FM4-B is automatically satisfied regardless of  $\gamma$  and  $S$  — cooperation is already the dominant strategy.

## 2.4 Participant Growth Rate $g(t)$ (Parameter 5.4)

## 2.5 Interaction Topology Modifier (Parameter 5.8)

Used to adjust the critical mass threshold  $N^*$ :

| User selection                       | Numerical value |
|--------------------------------------|-----------------|
| Stable ( $g \approx 0$ )             | $g = 0.00$      |
| Slow growth ( $g \approx 0.05$ )     | $g = 0.05$      |
| Moderate growth ( $g \approx 0.10$ ) | $g = 0.10$      |
| Fast growth ( $g \approx 0.20+$ )    | $g = 0.20$      |

| Topology             | Factor | Adjusted $N^*$                |
|----------------------|--------|-------------------------------|
| Well-mixed           | 1.00   | $N^* = 2Kd + 1$               |
| Spatially structured | 0.60   | $N^* = 0.60 \times (2Kd + 1)$ |
| Network-based        | 0.75   | $N^* = 0.75 \times (2Kd + 1)$ |

## 2.6 Burn Rate $B(t)$ (Parameters 3.1 and 3.2)

- **None:**  $B(t) = 0$
- **Demand-driven:**  $B(t) = Q \times \bar{P} \times f_b$ , where  $f_b$  is the burn fraction from sub-input (default  $f_b = 1.0$ )
- **Rule-driven:**  $B(t)$  = burn rate per period from sub-input
- **Threshold-driven:**  $B(t) = b_r$  if  $M(t) > M^*$ , else  $B(t) = 0$ , where  $M^*$  and  $b_r$  from sub-inputs
- **Expiry:**  $B(t) \approx M \times (1 - s_r)$ , where  $s_r = \min(1, d \times \bar{P}/B_{avg})$
- **Coupon-layer only:**  $B(t) = 0$  at token layer; flag as partial mechanism

## 3 Derived Variables

Compute in the following order to avoid dependency issues.

### 3.1 Average Token Price $\bar{P}$

- **Physical quantity:**  $\bar{P} = q_t \times p_u$  (quantity per token  $\times$  market price per unit)
- **Service or access unit:**  $\bar{P}$  = tokens per service unit (e.g. 1 token/day borrowing  $\Rightarrow \bar{P} = 1$ )
- **Market price:**  $\bar{P}$  = current market price from sub-input (flag as volatile)
- **Pegged to external currency:**  $\bar{P}$  = peg ratio from sub-input

- **None:**  $\bar{P} = 1$  (normalized unit; flag as simplification)
- **Specific goods/services with tiers:**  $\bar{P} = \sum_i w_i \times p_i$  (weighted average of coupon tier values  $p_i$  with weights  $w_i$ ; e.g. equal distribution of tiers [5, 10, 15] gives  $\bar{P} = 10$ )
- **Time-based borrowing:**  $\bar{P} = c_d \times \bar{l}$  (cost per day  $\times$  average loan duration)

### 3.2 Expected Transaction Volume $Q$

If the user enters  $Q$  directly (parameter 5.2), use the entered value. Otherwise compute:

$$Q_{supply} = \frac{N(N-1)}{2K} \times r_a \quad (3)$$

where  $r_a = 0.5$  is the default activity rate.

$$Q_{demand} = N \times d \quad (4)$$

Use the conservative estimate:

$$Q = \min(Q_{supply}, Q_{demand}) \quad (5)$$

### 3.3 Total Emission Rate $E(t)$

$$E(t) = f_a \times N \times Y \quad (6)$$

where  $f_a$  is activity frequency (parameter 5.3) and  $Y$  is the emission function output (parameter 2.2). For piecewise functions with multiple reward types:

$$E(t) = \sum_j f_{a,j} \times N \times Y_j \quad (7)$$

Add initial distribution amortized over 52 weeks:

$$E_{total}(t) = E(t) + \frac{D_{initial}}{52} \quad (8)$$

where  $D_{initial}$  is total initial tokens distributed.

### 3.4 Wealth-Weighted Average Holding Time $\bar{\tau}$

$$\bar{\tau} = \sum_i \frac{M_i}{M_{circ}} \times \tau_i \quad (9)$$

where  $M_{circ}$  excludes terminal wallets (actors with  $\tau \rightarrow \infty$ , e.g. merchant wallets). Convert holding times from days to periods:  $\tau_{periods} = \tau_{days}/7$ .

Flag actor types with  $\tau_i < 3$  days as high velocity risk subgroups.

### 3.5 Average Token Balance $B_{avg}$

$$B_{avg} = \frac{M_{total}}{N} \approx \frac{(E(t) - B(t)) \times 12}{N} \quad (10)$$

If  $M_{total} \leq 0$  (deflationary system): set  $B_{avg} = E(t) \times 2$ .

### 3.6 Consistent Token Supply $M_{consistent}$

$$M_{consistent} = \frac{\bar{P} \times Q}{V_{lower}} \quad (11)$$

where  $V_{lower} = 1/\bar{\tau}$  (computed after Equation 9).

*Note:* if  $V_{lower} = 0$  (tokens never spent),  $M_{consistent} \rightarrow \infty$ .

### 3.7 Active Contributor Rate $\phi$

Default estimates from verification mechanism (parameter 1.5):

| Verification mechanism | Default $\phi$ estimate |
|------------------------|-------------------------|
| Physical presence      | 0.80                    |
| Smart contract         | 0.75                    |
| Third-party cert       | 0.70                    |
| Peer verification      | 0.65                    |
| Self-reporting         | 0.40                    |

Override with user-provided value if available.

## 4 Failure Mode Calculations

### Multi-Token Systems

For systems with more than one token type (see Token Architecture sheet in the input specification), all failure mode calculations are applied *per token type* with role-specific threshold interpretations. The following rules apply before running any FM calculation:

1. Identify the token role from the Token Architecture input (utility / governance / reputation / resource)
2. Apply the FM applicability table to determine which conditions to evaluate for each token type
3. For **governance tokens**:

- Skip FM2 entirely — high  $\bar{\tau}$  is the design goal, not a risk
  - Relax FM3 —  $\rho = 0$  does not trigger a critical warning; suppress the burn mechanism absence warning
  - Add a token distribution check to FM6: if any single actor holds  $> 33\%$  of governance token supply, flag plutocratic capture risk regardless of  $\Gamma$
4. For **reputation tokens**: skip FM1, FM2, FM3, and FM5 entirely; apply FM4 and FM6 only
  5. For **resource tokens**: apply all FMs but note that FM1 risk is structurally reduced by the physical emission anchor; display this as a mitigating factor in the FM1 output
  6. The **overall risk score** (Section 6) is computed as the weighted average across all token types: utility token scores use weight  $\times 1.5$ ; governance token scores use weight  $\times 0.75$ ; reputation and resource token scores use weight  $\times 1.0$

Risk levels and colour codes used throughout:

- **GREEN** — condition satisfied (low risk)
- **AMBER** — borderline or moderate risk
- **RED** — condition violated (high/critical risk)

#### 4.1 FM1: Token Oversupply and Inflation Spiral

Formula.

$$r_{os} = \frac{E(t)}{M_{consistent}} = \frac{E(t) \times V_{lower}}{\bar{P} \times Q} \quad (12)$$

Risk levels.

**GREEN**  $r_{os} \leq 1.00$

*“Supply is within absorptive capacity. Token value is stable.”*

**AMBER**  $1.00 < r_{os} \leq 1.50$

*“Moderate oversupply risk. Emission rate exceeds absorptive capacity by  $[(r_{os} - 1) \times 100]\%$ . Recommend reducing emission or expanding redemption catalogue.”*



**RED**  $1.50 < r_{os} \leq 2.50$

*“High oversupply risk. System emits  $[r_{os}]x$  more tokens than the exchange market can absorb. Token value will erode over time without intervention.”*

**RED CRITICAL**  $r_{os} > 2.50$

*“Critical oversupply. Immediate redesign of emission rate or redemption structure required.”*

**Secondary check.** If  $V_{lower} > V^* = \bar{P}Q/M_{total}$ , append: *“Velocity is also above the stable threshold. High velocity is amplifying inflationary pressure — see FM2.”*

**Edge cases.**

- $Q = 0$ :  $r_{os} \rightarrow \infty \Rightarrow$  RED CRITICAL: *“No transaction volume. Tokens are emitted but cannot be redeemed.”*
- $E(t) = 0$ :  $r_{os} = 0 \Rightarrow$  GREEN
- $V_{lower} = 0$ :  $M_{consistent} \rightarrow \infty \Rightarrow$  GREEN (flag potential inactivity)

## 4.2 FM2: Token Velocity Trap

**Formulas.**

$$V_{lower} = \frac{1}{\bar{\tau}} \quad (13)$$

$$V^* = \frac{\bar{P} \times Q}{M_{total}} \quad (14)$$

By Jensen’s inequality, since  $f(x) = 1/x$  is convex:

$$V = \mathbb{E} \left[ \frac{1}{\tau} \right] \geq \frac{1}{\mathbb{E}[\tau]} = \frac{1}{\bar{\tau}} = V_{lower} \quad (15)$$

Therefore  $V_{lower}$  is a *lower bound* on true velocity. Always display this note to the user alongside the velocity estimate.

**Risk levels.**

**GREEN**  $\bar{\tau} > 14$  days AND  $V_{lower} \leq V^*$

*“Tokens are held for sufficient time. Velocity is within the stable range.”*

**GREEN borderline**  $7 < \bar{\tau} \leq 14$  days

*“Moderate holding time. Monitor fast-spending subgroups.”*

**AMBER**  $3 < \bar{\tau} \leq 7$  days AND  $V_{lower} \leq V^*$

*“Short holding times detected.  $V_{lower} = [\text{value}]$  per period. Consider adding holding incentives.”*

**RED**  $\bar{\tau} \leq 3$  days OR  $V_{lower} > V^*$

*“Velocity trap risk. Tokens are being spent almost immediately ( $\bar{\tau} = [\text{value}]$  days). True velocity is likely higher than  $[V_{lower}]$  due to fast-spending subgroups (Jensen’s inequality). Recommend: tiered redemption, governance rights, or time-locked rewards.”*

**RED CRITICAL**  $\bar{\tau} \leq 1$  day

*“Critical velocity trap. No holding incentive whatsoever. All tokens circulate at maximum speed, maximally amplifying any supply imbalance from FM1.”*

**Complementary currency note.** If NFR6 = “circulate fast”, replace RED with AMBER and append: *“Note: high velocity is consistent with your declared circulation intent. Velocity trap risk applies only if  $V > V^*$ .”*

**Edge cases.**

- $\bar{\tau} \rightarrow \infty$  (all tokens terminal):  $V_{lower} = 0 \Rightarrow$  GREEN for velocity; flag system inactivity risk separately
- $M_{total} = 0$ :  $V^*$  undefined; skip FM2 and flag pre-launch state

### 4.3 FM3: Burn/Emission Imbalance

**Formulas.**

$$\rho = \frac{B(t)}{E(t)} \quad (16)$$

$$\Delta M = E(t) - B(t) \quad (17)$$

Full sustainability condition:

$$E(t) - B(t) \leq g(t) \cdot M(t) \quad (18)$$

**Risk levels.**

**GREEN**  $\rho \geq 1.0$  AND Equation 18 satisfied

*“Burn mechanism covers emission. Supply is stable or deflationary.  $\rho = [\text{value}]$ .”*

**AMBER**  $0.50 \leq \rho < 1.0$

*“Partial burn coverage. Supply grows at  $[\Delta M]$  tokens/period. Burn covers  $[\rho \times 100]\%$  of emission. Acceptable during growth phase but unsustainable when stable.”*

**RED**  $0 < \rho < 0.50$

*“Insufficient burn coverage. Supply grows at  $[\Delta M]$  tokens/period. Burn covers only  $[\rho \times 100]\%$  of emission.”*

**RED CRITICAL**  $\rho = 0$  (no burn mechanism)

*“No burn mechanism. Token supply grows monotonically at  $[E(t)]$  tokens/period with no corrective mechanism. This is the most common structural vulnerability in community token economies. Recommend demand-driven burn triggered at point of redemption.”*

#### Burn type quality flags.

- Rule-driven burn: append AMBER flag — *“Does not respond to demand dynamics.”*
- Coupon-layer only: append AMBER flag — *“Token supply remains uncorrected.”*
- Expiry: append note about panic spending risk

#### Edge cases.

- $E(t) = 0$ :  $\rho$  undefined; set  $\rho = 1 \Rightarrow$  GREEN
- $B(t) > E(t)$ :  $\rho > 1$ , deflationary  $\Rightarrow$  GREEN with note: *“Monitor for token scarcity.”*

## 4.4 FM4: Free-Rider and Contribution Collapse

Two conditions. **Condition A — Contribution balance:**

$$\phi_{\text{required}} = \frac{d}{K} \quad (19)$$

$$\text{Condition A} = (\hat{\phi} \geq \phi_{\text{required}}) \quad (20)$$

**Condition B — Monitoring deterrence:**

$$\text{Condition B} = (\gamma S > T - R) \quad (21)$$

#### Risk levels.

**GREEN** Both A and B satisfied

*“Contribution balance maintained. Monitoring and sanctions are sufficient to deter free-riding.”*

**AMBER** A satisfied, B not satisfied

*“Contributor rate is adequate but monitoring is insufficient.  $\gamma S = [\text{value}] \leq T - R = [\text{value}]$ . Recommend increasing  $\gamma$  or  $S$ .”*

**AMBER** B satisfied, A not satisfied

*“Monitoring is adequate but contributor rate may be insufficient. Required:  $\phi \geq [\text{value}]$ . Estimated:  $\hat{\phi} = [\text{value}]$ .”*

**RED** Neither A nor B satisfied

*“High free-rider risk. Both contribution rate and monitoring conditions are violated.”*

**RED CRITICAL**  $\phi_{\text{required}} > 0.80$

*“Required contributor rate exceeds 80% — not sustainable in practice. Reduce  $d$  or increase  $K$ .”*

**Actor-specific check.** For each Provider actor type: if verification mechanism does not include smart contract automation, append AMBER: *“Provider compliance is unmonitored. Provider free-riding is the most common failure point in community token economies.”*

**Edge cases.**

- $K = 0$ : configuration error; request  $K \geq 1$
- $d = 0$ :  $\phi_{\text{required}} = 0 \Rightarrow \text{GREEN}$
- $T - R \leq 0$ : Condition B automatically satisfied

## 4.5 FM5: Insufficient Critical Mass

**Formula.**

$$N^* = f_{\text{topo}} \times (2Kd + 1) \quad (22)$$

where  $f_{\text{topo}}$  is the topology modifier from Section 2.5.

$$r_{\text{cm}} = \frac{N}{N^*} \quad (23)$$

### Risk levels.

**GREEN**  $r_{cm} \geq 2.0$

*“Participant count is well above critical mass threshold.  $N = [N]$ ,  $N^* = [N^*]$ .”*

**GREEN borderline**  $1.20 \leq r_{cm} < 2.0$

*“Exceeds critical mass with limited margin.”*

**AMBER**  $1.00 \leq r_{cm} < 1.20$

*“Just above critical mass threshold. A loss of  $[N - N^*]$  participants would push the system below viability.”*

**RED**  $r_{cm} < 1.00$

*“Below critical mass. System needs  $[N^* - N]$  more participants.”*

**RED CRITICAL**  $r_{cm} < 0.50$

*“Critical mass severely insufficient. System needs more than double current participants.”*

**Supply-side bottleneck check.** If  $Q_{supply} < Q_{demand}$  (even when  $N \geq N^*$ ), append AMBER: *“Demand exceeds supply-side capacity. Expand provider network or increase offer renewal frequency.”*

### Edge cases.

- $K = 0$ : configuration error
- $d = 0$ :  $N^* = 1 \Rightarrow$  GREEN for any  $N \geq 1$
- $N = 0$ : RED CRITICAL

## 4.6 FM6: Governance Capture

### Formula.

$$\Gamma = \frac{|\text{single entity decisions}|}{|\text{total governance decisions}|} \quad (24)$$

Total governance decisions = 7 (fixed list from parameter 4.2): emission rate; burn rate; eligibility criteria; provider onboarding; exchange ratios; reward structure; rule modification.

### Risk levels.

**GREEN**  $\Gamma \leq 0.30$

*“Governance is well distributed.”*

**GREEN borderline**  $0.30 < \Gamma \leq 0.50$

*“Governance is moderately distributed. Approaching the centralization threshold.”*

**AMBER**  $0.50 < \Gamma \leq 0.80$

*“Governance significantly centralized.  $\Gamma = [\text{value}]$ . Recommend distributing reward structure and eligibility decisions.”*

**RED**  $\Gamma > 0.80$

*“High governance centralization.  $\Gamma = [\text{value}]$ . System is structurally dependent on a single entity.”*

**Bootstrapping note.** Always append for  $\Gamma > 0.5$ : *“Centralization at launch is expected in community token economies. The risk is the absence of a transition pathway. [If roadmap defined: ‘A decentralization roadmap has been defined — this partially mitigates the risk.’] [If no roadmap: ‘No decentralization roadmap defined. Recommend specifying governance delegation milestones.’]”*

**Roadmap modifier.** If governance type = Hybrid AND transition milestones are defined: reduce effective risk by one level (RED  $\rightarrow$  AMBER; AMBER  $\rightarrow$  GREEN borderline).

## 5 Coherence Check Logic

Run after all six failure modes are computed. Each contradiction generates a warning in the coherence report section.

### C1. Velocity vs Circulation Intent

*Condition:* NFR6 = “retain value” AND holding incentive = None only AND  $\bar{\tau} < 7$  days

*Warning:* “Declared intent to retain value requires holding incentives, but none are defined.”

### C2. Proportionality vs Verification (CRITICAL)

*Condition:* NFR5  $\geq 4$  AND verification = Self-reporting

*Warning:* “Self-reporting cannot guarantee proportional rewards. Strongest free-rider collapse risk.”

### C3. Resilience vs Burn Mechanism (CRITICAL)

*Condition:*  $\text{NFR1} \geq 4$  AND burn mechanism = None

*Warning:* “No burn mechanism is the most common structural vulnerability in community token economies.”

### C4. Governance Intent vs Design

*Condition:*  $\text{NFR7} = \text{“immediate”}$  AND  $\Gamma = 1.0$  AND no roadmap defined

*Warning:* “Governance maturity intent inconsistent with fully centralized design.”

### C5. DAO Governance vs Accessibility

*Condition:*  $\text{NFR3} \geq 4$  AND governance type = DAO-governed

*Warning:* “Token-weighted voting may exclude non-technical participants.”

### C6. Expiry vs Retain Value

*Condition:*  $\text{NFR6} = \text{“retain value”}$  AND burn mechanism = Expiry

*Warning:* “Token expiry penalizes holders, contradicting retain value intent.”

### C7. Reinforcing Supply-Side Triad (CRITICAL)

*Condition:*  $r_{os} > 1.5$  AND  $\rho = 0$  AND  $\Gamma = 1.0$  with no roadmap

*Warning:* “System has three simultaneous supply-side vulnerabilities: oversupply, no burn mechanism, and no governance mechanism to correct them. This is the highest-risk configuration in community token economy design.”

## 6 Overall Risk Score

### 6.1 Individual Failure Mode Scores

| Risk level        | Score |
|-------------------|-------|
| GREEN (satisfied) | 0     |
| GREEN borderline  | 1     |
| AMBER             | 2     |
| RED               | 3     |
| RED CRITICAL      | 4     |

### 6.2 Weights

### 6.3 Total Score and Interpretation

$$S_{weighted} = \sum_i (s_i \times w_i) + \sum_j p_j \quad (25)$$

| FM                             | Name                       | Weight    |
|--------------------------------|----------------------------|-----------|
| FM1                            | Token Oversupply           | 1.5       |
| FM2                            | Velocity Trap              | 1.0       |
| FM3                            | Burn/Emission Imbalance    | 1.5       |
| FM4                            | Free-Rider Collapse        | 1.0       |
| FM5                            | Insufficient Critical Mass | 1.0       |
| FM6                            | Governance Capture         | 0.5       |
| Critical contradiction penalty |                            | +1.0 each |
| Standard contradiction penalty |                            | +0.5 each |

$$S_{norm} = \frac{S_{weighted}}{S_{max}} \times 100 \quad (26)$$

where  $S_{max} \approx 33$  (all RED CRITICAL plus all contradictions).

| Range   | Level         | Message                                                                                                     |
|---------|---------------|-------------------------------------------------------------------------------------------------------------|
| 0–20%   | LOW RISK      | “The design is broadly sound. Minor optimisations recommended.”                                             |
| 21–40%  | MODERATE RISK | “Several conditions require attention before deployment.”                                                   |
| 41–60%  | HIGH RISK     | “Significant structural vulnerabilities. Deployment not recommended without addressing flagged conditions.” |
| 61–100% | CRITICAL RISK | “Multiple critical failure modes. Major redesign required.”                                                 |

## 6.4 Priority Recommendation List

After scoring, generate a prioritised list sorted by:

1. RED CRITICAL failure modes
2. Critical coherence contradictions
3. RED failure modes
4. AMBER failure modes
5. Standard coherence contradictions



For each item display: which failure mode or contradiction; the specific condition violated; the concrete design change that resolves it; the estimated improvement in risk level if applied.

## 7 Developer Notes

### 7.1 Computation Order

To avoid dependency issues, compute in this order:

1. Pre-process all qualitative inputs (Section 2)
2. Compute  $\bar{P}$  (Section 3.1)
3. Compute  $Q$  (Section 3.2)
4. Compute  $E(t)$  (Section 3.3)
5. Compute  $\bar{\tau}$  (Section 3.4)
6. Compute  $V_{lower} = 1/\bar{\tau}$  and  $V^* = \bar{P}Q/M_{total}$
7. Compute  $B_{avg}$  (Section 3.5)
8. Compute  $M_{consistent}$  (Section 3.6)
9. Compute  $\hat{\phi}$  (Section 3.7)
10. Compute  $T$  and  $R$  (Section 2.3)
11. Compute  $B(t)$  (Section 2.6)
12. Compute  $\rho = B(t)/E(t)$
13. Compute  $\Gamma$  (FM6)
14. Run FM1–FM6 assessments
15. Run coherence checks
16. Compute overall score

## 7.2 Unit Conventions

All monetary values are expressed in tokens for internal calculations. Currency values from sub-inputs are converted to token equivalents using  $\bar{P}$  before any formula is applied.

All time values are converted to weeks:

- Days to weeks: divide by 7
- Months to weeks: multiply by 4.33

## 7.3 Precision

Intermediate calculations: 4 decimal places. Display to user: 2 decimal places.

## 7.4 Missing Inputs

If a required input is missing and no default is defined, flag the input as required and block the diagnostic calculation for the affected failure mode only. Other failure modes should still compute if their inputs are complete.